# Know your little lab.

## Print the enclosure. Build the sensor. Notice room changes.




I built LabSense for a little lab where printers, resin work and a workbench share the same room. We spend hours improving the machines; the air around them deserves attention too. This small USB-powered project keeps a VOC estimate, temperature and humidity visible, with local Home Assistant history if wanted.

**Working prototype / firmware 1.3.0-rc6.** Physical measurement acceptance and replacement-module fit are still in progress. This guide covers LabSense Original; it is not a certified instrument or an approved stable software release.

## What the sensor can tell you

CCS811 estimates broad VOC behavior. It does not identify resin chemicals, measure ultrafine printer particles, directly measure CO2 or certify safe air. Keep suitable extraction, ventilation and material safety practices regardless of a low reading. Warning thresholds are technical indicators, not health or legal exposure limits.

Design sketches are illustrations. The supplied workshop photographs show the actual prototype. Neither type of image establishes calibrated measurement accuracy.

# Why monitor the room?

A modest DIY project can make changing room conditions easier to notice. The useful part is the everyday habit: look at the screen, review a trend, and investigate what changed. LabSense Original complements a well-managed workshop; it does not establish that the air is safe.

## VOC: three letters, many different chemicals

**VOC means volatile organic compound.** These are carbon-based chemicals that can enter room air as gases. Some can irritate eyes or airways; effects depend on the chemical and exposure. A VOC estimate cannot tell you the composition of a mixture.

In a printing room, possible sources include heated filament, resin work, alcohol used to wash prints, adhesives and cleaning products. FDM/FFF and resin processes can also release particles. Emissions vary with materials, operating conditions and ventilation. Preparation, washing, curing and cleaning deserve attention alongside the print itself.

## What the original hardware can tell you

| Reading | Useful interpretation | Important limit |
|---|---|---|
| VOC estimate / ppb | CCS811 broad gas response processed into equivalent total VOC (eTVOC). Alcohol vapours and other sources may affect it. | Cannot identify a chemical, attribute its source or quantify a named resin ingredient. |
| Temperature / relative humidity | BME280 readings at the device location; also used for gas compensation. | Device heat and placement matter. Not calibrated complete-room accuracy or filament moisture. |
| Home Assistant trends | Review ordinary work and ventilation periods; create your own reminders for a sustained rise. | No automation is installed. Correlation does not prove a source or the adequacy of ventilation. |

**ppb = parts per billion; 1 ppm = 1,000 ppb.** Here it is a gas-equivalent algorithmic scale, not a laboratory sum of every VOC. The optional eCO2 entity is VOC-derived, not a direct CO2 measurement.

## Use it for awareness, keep emission controls in place

This hardware has no particle sensor and does not directly measure CO2 or CO. It is not a resin detector, smoke detector or safety alarm. A low reading or a falling trend does not certify safe air. Keep appropriate extraction, ventilation and material handling in place; do not wait for a warning or deliberately release fumes to test it.

Further reading: LabSense practical guide; EPA VOC overview; NIOSH 3D-printing guidance; ams CCS811 datasheet.

# Source parts that fit.

Research snapshot: 16 October 2026, EUR. Match signal order and mechanical stack before ordering multiples. Similar-looking breakouts can have incompatible pin orders. Supplier candidates below have not been physically approved as replacements.

| Qty / part | Required variant / source | Indicative cost |
|---|---|---|
| 1 XIAO ESP32C3 | Seeed 113991054 / BerryBase SE-113991054; 4 MB, 2x7 pins; antenna usually included | 7.90 incl. VAT |
| 1 CCS811 | BerryBase CJ-CCS811; eight-pin CJMCU-style breakout | 11.80 incl. VAT |
| 1 BME280 | HESTORE 100.390.68; four pins VIN-GND-SCL-SDA; reject BMP280 | 5.96* |
| 1 SSD1306 OLED | HESTORE 100.483.04 GV-WH; 0.96 inch, 128x64 I2C; GND-VCC-SCL-SDA | 3.07* |
| 1 bare carrier | LabSense original Gerbers; 55x50x1.5 mm, 2-layer FR-4 | 1-3 allocated |
| 30 male + 30 female contacts | 2.54 mm: 2x7, 1x8, 2x4; buy cutting allowance; height matters | 1-2 |
| 3 case screws | M3x12 cylinder/button heads; no countersunk heads | 0.30-1 |
| 3 printed parts | Base, lid, wall plate; PETG | 1-2 material |

* HESTORE observations: EUR 5.008 and 2.576 net, converted with illustrative 19% VAT. Destination checkout prices differ. All delivery is extra. See the GitHub BOM CSV for clickable shop links.

**Materials about EUR 32-37 per unit** using an allocated PCB and existing USB accessories. A first unit paying for a whole small PCB batch is about EUR 36-49 before delivery and pack minimums. PCB figures are planning allowances, not a quote. Add a USB data cable (allow EUR 3-6) and 5 V supply (EUR 5-10) if needed. Tools, solder, wall fixings, labor and failed prints are excluded.

## Order bare PCBs with the original Gerber ZIP

Download the original Gerber ZIP from hardware/pcb/source in the GitHub project. Upload only that fabrication archive unchanged to PCBWay Quick Order. Choose single-design bare FR-4, two layers, 55x50 mm and **1.5 mm thickness**. Ask the fabricator if its standard form only offers 1.6 mm. Mask color is your choice. Standard 1 oz copper and lead-free HASL are budget reorder suggestions, not recovered invoice data.

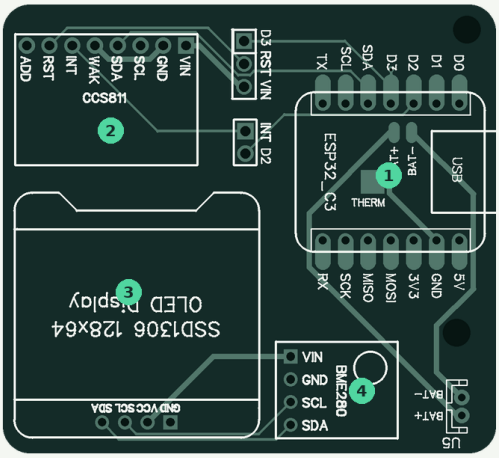
Review GKO outline, copper/mask/silkscreen mapping and plated/unplated drills. GDL is documentation, not extra copper or a second outline. Resolve fabrication questions, including any near-zero-length slot flags, without deleting source drill data. Confirm the full shipping/tax quote before payment. This order supplies bare carriers, not populated electronics.

# Know every pin.

LABSENSE / BUILD GUIDE

## Four modules. One carrier.

Component-side assembly map / USB to the right / compare every delivered pin label




1. **XIAO ESP32C3**
   2 × 7 pins / USB at the board edge
   SDA D4 · GPIO6 / SCL D5 · GPIO7
2. **CCS811 VOC module**
   8 pins / match VCC through ADD
   Sensor opening faces room air
3. **SSD1306 OLED**
   4 pins / GND · VCC · SCL · SDA
   Reject the reversed-power variant
4. **BME280 climate module**
   4 pins / VIN · GND · SCL · SDA
   BMP280 cannot measure humidity

Pin rows come from Gerbers; numbered module locations are assembly annotations.

GERBER-DERIVED / ORIGINAL CARRIER / MILLIMETERS

Illustration only. Original Gerbers remain the fabrication source. Colors are illustrative.

Gerber-derived illustration of the component side, USB to the right. Use the carrier signal labels; do not match only the color or apparent orientation of a shop photograph. All breakout supplies on this carrier are 3.3 V.

| Module / location | Carrier sequence in the view above |
|---|---|
| XIAO / upper right | Upper row left to right: TX, SCL, SDA, D3, D2, D1, D0.<br>Lower row left to right: RX, SCK, MISO, MOSI, 3V3, GND, 5V. |
| CCS811 / upper left | Left to right: ADD, RST, INT, WAK, SDA, SCL, GND, VCC (carrier supply label VIN). |
| OLED / lower left | Left to right at lower edge: SDA, SCL, VCC, GND.<br>From GND end: **GND-VCC-SCL-SDA**. |
| BME280 / lower right | Along left edge, top to bottom: **VIN-GND-SCL-SDA**. |

**I2C:** XIAO D4/GPIO6 = SDA, D5/GPIO7 = SCL, 100 kHz. CCS811 WAK and ADD are grounded; address 0x5A. INT/RST are exposed at auxiliary headers but unused by this firmware. OLED supports 0x3C/0x3D; BME280 supports 0x76/0x77.

**Reject direct-insertion substitutes with swapped power pins.** OLED VCC-GND-SCL-SDA is incompatible. BMP280 cannot measure humidity. Larger/six-pin sensors, SPI displays, 128x32 OLEDs and SH1106 modules are not this build. Leave battery/THERM and auxiliary headers unused for the USB-only build.

# Solder, inspect, measure.

Tools: suitable soldering iron, electronics solder/flux, holder, cutters, magnifier, multimeter, calipers and M3 screwdriver. Use soldering ventilation and eye protection. Breakout chips are already assembled; only headers/sockets need soldering.

## Build the removable module stack

1. Disconnect USB. Dry-fit all modules and sockets. Female sockets sit on the carrier component side, tails through the board. Module male pins point down into them. XIAO row spacing is 15.24 mm.

2. Cut sockets/headers to 2x7, 1x8 and 2x4. Use an unpowered mating strip/jig to align them. Tack one end, check seating and parallel rows, tack the opposite end, then finish the joints.

3. Solder carrier sockets from below. Solder module headers from their appropriate pad face. Keep the iron away from glass, flex cables, antenna connector and sensor ports.

4. Inspect every joint for wetting to both pin and pad. Repair bridges and incomplete joints. Remove loose solder debris. Do not spray or immerse sensor openings in cleaner; follow flux handling instructions.

5. With power disconnected, check GND/3V3/SDA/SCL continuity to every module location and no persistent near-zero rail short. A brief capacitive meter indication is not by itself a short.

6. Fit the XIAO and external antenna first. Power only this assembly and carefully check about 3.3 V relative to GND at each breakout supply pad. Disconnect USB before inserting the remaining checked modules.

## The enclosure depends on stack height

| Datum: carrier underside = zero | Measured prototype / required clearance |
|---|---|
| Carrier thickness | 1.5 mm |
| Case floor to carrier underside | 4 mm for solder tails |
| Highest point / display face | 9.5 mm above underside |
| USB-C bottom / top | 5.2 / 8.9 mm above underside |
| USB-C protrusion past carrier | 0.3 mm |
| OLED glass | 26.7x19.6 mm; confirm position over opening |

A different socket height moves both USB and glass. Check fit before soldering permanently. Trim only excess solder tails where required; keep mating engagement intact. Never use the lid to press a taller display into place.

# Print. Fit. Slide into place.




Actual v3 CAD geometry; exploded spacing is illustrative. Electronics and screws are omitted. Get the 3MF and print profile from MakerWorld; enclosure models are distributed only there under SDFL. For PETG, the starting point is H2D, 0.4 mm nozzle, 0.16 mm quality, Arachne and automatic supports. Inspect the supplied profile and check printed fit; these suggested settings are not an independently validated profile.

**Orient and clean.** Base rear face to bed; lid smooth outer face to bed; wall plate flat wall face to bed. Remove supports without damaging slide rails, vents, screw pockets or the thin display collar. Check the first layer and material/plate pairing in the slicer.

**Fit unpowered.** Seat the foil antenna below the carrier, electrically insulated from solder tails. Avoid pinching its lead or pulling its small connector. Keep all sensor ports and vents clear. The antenna location still needs reception checks at the actual installation.

**Close gently.** Seat the board and check USB access and glass clearance. Use three M3x12 cylinder/button-head screws, measured under the head; no countersunk screws. Heads should fit the nominal 6.4 mm diameter x 3.2 mm pockets. Lid pilots are 2.5 mm. Form threads carefully and tighten lightly by hand.

**Mount.** Use suitable screws/anchors or adhesive for the actual wall. Attach the plate, slide the enclosure down onto it and leave clearance above for removal. The slide has no anti-lift lock. It is a ventilated indoor case with no water/dust rating.

Nominal v3 CAD case envelope: 66.5x57x19.8 mm; mounted depth 22.5 mm excluding adhesive/wall unevenness. Wall plate outline 64.5x55 mm, inset 1 mm per edge. Real printed fit, replacement modules and closed-case temperature bias still need checks.

# ESPHome, from source to Wi-Fi.

LabSense firmware is built with **ESPHome**, using its native encrypted local API for Home Assistant. You do not need an MQTT broker or cloud service for sensor data. A Home Assistant installation is optional for the local OLED, but required for its dashboards/history/automations.

## Build the included source on a fresh controller

Get the project from github.com/zrnnn/LabSense. Use Code > Download ZIP, extract it and open a terminal in the project folder (or clone it with Git). The repository contains the complete firmware/release inputs, fonts, custom components, offline setup assets, Gerbers and guides. Enclosure 3D files are available only on MakerWorld. Use Python 3.12 or newer, a USB data cable and an activated virtual environment. Firmware is a release candidate; no stable one-click binary is approved.

```
python -m venv .venv
# Activate .venv for your operating system
python -m pip install -r requirements.txt
python tools/build_portal_assets.py --variant release --check
python -m esphome compile firmware/release/labsense.yaml
python -m esphome upload firmware/release/labsense.yaml
# Choose the connected XIAO USB serial port
```

The generic release YAML needs no home Wi-Fi secrets or shared pairing key. Each fresh device generates and stores its own setup password and encryption key. Do not flash example-secret validation images, use the legacy stable YAML for a fresh build, or erase a paired device as an ordinary update. Serial installation uses a factory image; an OTA application must not be written at flash address zero. The GitHub project includes the quick start, wiring guide and license map; use this PDF for assembly. A direct ESPHome build from the downloaded project does not require Git metadata. Internet is needed for the initial toolchain/dependency download; device setup and operation stay local.

## Phone setup: save the key first

1. Power up. The OLED shows startup checks and the LabSense hotspot name/suffix and its eight-digit password. Join it on your phone; remain connected if told it has no internet.

2. Open the setup prompt or http://192.168.4.1. Press **Copy key** or **Save image**. This is the individual Home Assistant encryption key, not a Wi-Fi password. Keep it privately.

3. Choose your 2.4 GHz home Wi-Fi, enter its password and press Connect. Refresh starts another network search; Other network supports manual/hidden SSIDs. Display settings chooses the USB-port position and saves immediately.

4. Reconnect the phone to home Wi-Fi. In Home Assistant, open **Settings > Devices & services** and add the discovered ESPHome device using the saved key. If discovery fails, add its router-assigned address on port 6053.

**BLE is an alternative provisioning path, not normal iPhone Bluetooth pairing.** Save the key through the portal first. Then reconnect the phone to home Wi-Fi and use Home Assistant Improv with Bluetooth coverage, or a supported Android/computer browser at improv-wifi.com. Standard browser Improv is unsupported on iOS; the Wi-Fi portal is the direct iPhone route.

# Readings, warm-up and recovery.




Firmware-rendered OLED previews with fictional readings. Startup checks show sensor recognition and connection state; the running view remains fixed. It does not alternate pages. Four orientations support USB left/bottom/right/top.

| Condition | Expected behavior |
|---|---|
| Fresh generic installation | 48 continuous hours of valid initial gas sampling. An interrupted incomplete period restarts; partial hours are not saved. |
| Later restart after conditioning saved | 20 minutes of warm-up; baseline restore is attempted afterward. An extra 60 seconds follows successful restoration. |
| Temperature / humidity | Report independently. Allow thermal settling and compare the closed enclosure with a reference before applying offsets. |
| Fault / stale or uncompensated gas data | VOC assessment unavailable. OLED dashes/status and diagnostic entities identify the problem. Raw API values can exist during warm-up; check VOC evaluation ready. |
| High VOC estimate | Default 2,000 ppb technical indication: immediate valid-ready OLED warning; Home Assistant warning after 30 seconds. Not a health limit. |

**Warm-up is not calibration.** The 48-hour initial policy is a conservative LabSense choice. No independently validated product accuracy, compound identification, particle measurement or direct CO2 measurement is claimed. Use ordinary room observations; do not create hazardous emissions for testing. Home Assistant notifications need your own automation; this firmware does not install one.

**Recover setup:** while already running, hold BOOT for eight seconds, then release at the prompt. Recovery clears Wi-Fi and keeps the pairing key. Open the case if necessary. BOOT held during power-up instead enters the USB bootloader. Factory erase clears identity and conditioning.

**First checks:** blank OLED: power-pin order/address; missing humidity: genuine BME280 (ID 0x60); VOC fault: CCS811 pin order/WAK/ADD (ID 0x81, address 0x5A); key request: use saved API key; high temperature: allow settling and check location/ventilation. Sensor IDs do not prove calibrated accuracy.

# References and review checklist.

## Sources and useful links

**NIOSH: Safe 3D Printing**
https://www.cdc.gov/niosh/bulletin/2024/safe-3d-printing.html

**ESPHome documentation**
https://esphome.io/

**Home Assistant ESPHome integration**
https://www.home-assistant.io/integrations/esphome/

**Seeed XIAO ESP32C3 getting started**
https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/

**PCBWay bare PCB order**
https://www.pcbway.com/QuickOrderOnline.aspx

**XIAO / BerryBase**
https://www.berrybase.de/seeed-xiao-esp32c3-winziges-mcu-board-mit-wlan-und-ble

**CCS811 / BerryBase**
https://www.berrybase.de/ccs811-luftqualitaets-sensor-mit-breakout-board

**BME280 / HESTORE**
https://www.hestore.eu/en/prod_10039068.html

**OLED GV-WH / HESTORE**
https://www.hestore.eu/en/prod_10048304.html

## Before calling a build finished

Record module variants and socket heights. Check the power rail, sensor recognition, first setup, saved-key pairing, recovery, warm-up completion, display orientation, USB clearance, screw fit, wall-slide retention and actual antenna reception. Compare closed-case temperature/RH with a suitable reference. Photograph the actual printed profile result; a slicer preview does not prove printed fit.

## Current release limits

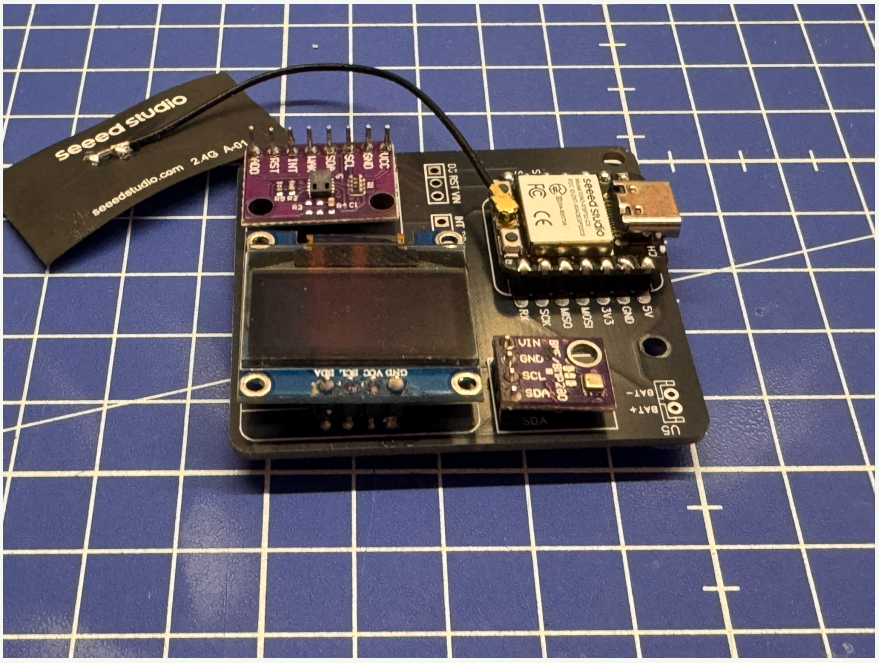
Firmware is 1.3.0-rc6. The owner has supplied photographs of a working printed assembly. Full Home Assistant pairing acceptance, reference measurements, candidate print-profile validation and other product release gates are not complete. The MakerWorld model listing is awaiting publication. The owner manages the uploaded print profile; this guide does not certify its specific material or settings.

## Reuse and attribution

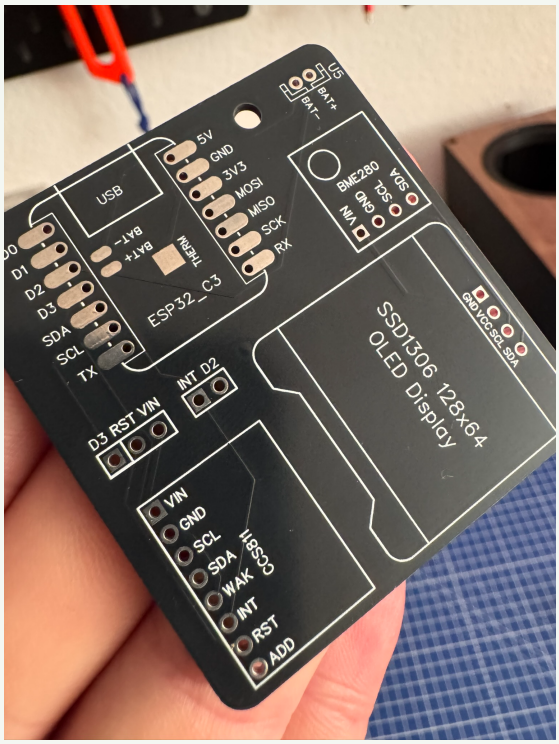
Original LabSense software: MIT. PCB hardware: CERN-OHL-P-2.0. Enclosure models: MakerWorld Standard Digital File License, distributed only on MakerWorld. Documentation and original illustrations: CC BY 4.0. Preserve GitHub license files and third-party notices; the model license does not replace software/PCB licenses. Cover/context sketches are design illustrations. PCB art derives from the original Gerbers; OLED art is rendered from firmware with fictional data. Real prototype photographs are supplied by the project owner. No Wi-Fi passwords, pairing keys, personal paths or household telemetry belong in shared files.

# From bare carrier to modules.

Real prototype photographs supplied by the project owner. The bare carrier is identified by the owner as a JLCPCB delivery. The earlier PCBWay ordering route is an alternative fabrication workflow; the photo is not proof of a particular PCBWay order or finish.




Populated carrier with XIAO ESP32C3, external foil antenna, CCS811, BME280 and SSD1306 OLED. Identify modules by signal labels, not board color. Socket/module height must reproduce the measured stack.



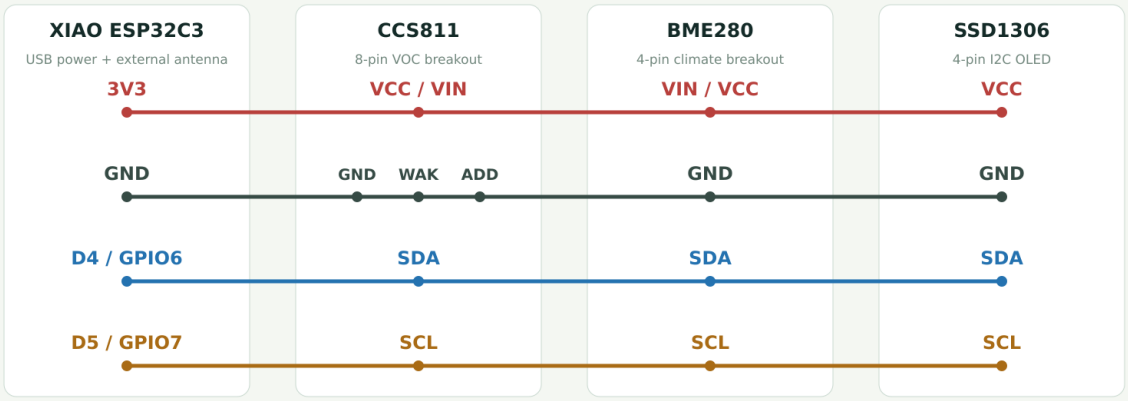

Bare carrier before soldering. Fabrication supplies the board; controller, sensors, OLED and headers/sockets are separate components. Check every pin against the assembly map before applying power.

# No carrier? Wire the same bus.

LABSENSE / CARRIER-FREE WIRING

## Four shared nets. The same firmware.

Breadboard, perfboard or your own carrier - connect by signal label, not physical position.




**Dots on the same horizontal line are connected. WAK + ADD must also go to GND.**

3.3 V modules and pull-ups only. Do not connect breakout supplies or SDA/SCL to 5 V.

CCS811: 0x5A. Leave INT unused; RST must remain high/inactive (check the breakout pull-up).

OLED: 0x3C/0x3D. BME280: 0x76/0x77. GPIO6 = SDA; GPIO7 = SCL. Bus speed: 100 kHz.

If I2C pull-ups are absent: one 4.7k resistor SDA to 3V3, one 4.7k SCL to 3V3.

Existing breakout pull-ups count in parallel. Power off before wiring; verify rails before fitting modules.

Signal diagram, not physical pin order. Original-carrier reconstruction; wired variant needs its own fit/measurement checks.

A breadboard, perfboard or your own carrier can use the same XIAO ESP32C3 and three 3.3 V breakout modules with the release firmware unchanged. The drawing groups electrical nets; it is not physical pin order. Match the delivered labels. This wiring is reconstructed from the original carrier and has not been separately bench-validated.

**CCS811:** WAK/nWAKE and ADD/ADDR must go to GND (awake, address 0x5A). Leave INT unused. RST is not firmware-controlled: verify the breakout holds it inactive/high; if its board lacks a pull-up, provide one per its schematic (commonly 10k to 3V3). Never ground RST.

**I2C:** keep short leads and the 100 kHz firmware setting. Breakouts often have pull-ups; count those already fitted. If absent, start with one 4.7k SDA-to-3V3 resistor and one 4.7k SCL-to-3V3 resistor. Do not use 5 V pull-ups. Bridge any split breadboard rails and continuity-check the unpowered circuit.

**First power:** verify rails with the XIAO alone, disconnect before adding modules, then follow the same source build, setup and conditioning steps. The current enclosure fits the original carrier and measured component positions; an arbitrary breadboard/perfboard will need its own support or case. Allow EUR 3-8 for board/wire as an unquoted alternative-build allowance.

Full instructions on GitHub: docs/wiring.md; diagram: docs/images/product/05-carrier-free-wiring.png; setup: docs/installation.md. Reference pin mapping: Seeed XIAO ESP32C3 guide. CCS811 wiring reference: github.com/sciosense/CCS811_driver. Its generic ESP32 examples use different GPIOs; LabSense uses GPIO6/GPIO7. CCS811 is discontinued; this reproduces the existing prototype, not a new sensor recommendation.